

Incident Report KR-INC-2026-003

T-07 gearbox bearing failure – Kestrel Ridge Wind Farm

David Chen · 21 January 2026 · KR-INC-2026-003

Field	Details
Incident number	KR-INC-2026-003
Category	Asset damage – major component failure (no injury, no environmental release)
Date / time	14 January 2026, 03:42 (turbine trip)
Location	Kestrel Ridge Wind Farm, turbine T-07 (Craigmuir Estate)
Reported by	Weyland Control Centre, Edinburgh (night shift)
Severity	Commercial: high · HSE: low
Status	Closed for operations on 6 February 2026; root cause analysis and warranty claim ongoing

1. What happened

At 03:42 on 14 January 2026, turbine T-07 tripped on the alarm *Gearbox ISS bearing temperature high* (set point 95 °C, measured 97.4 °C). In the hour before the trip, the bearing temperature had risen from 71 °C to 97 °C while the turbine was running at close to rated power (mean wind speed 13.8 m/s). The control room operator followed the alarm response matrix: no remote reset was attempted, the turbine was left stopped and Highline Wind Services was called out for the morning.

Stuart Laidlaw and a second technician climbed T-07 at 09:15. They found metal particles on the magnetic plug of the gearbox and a strong smell of overheated oil in the nacelle. An endoscope inspection through the inspection port showed severe spalling on the inner ring of the rotor-side bearing of the intermediate-speed shaft (ISS). The turbine was locked out pending a decision on repair.

IMAGE PLACEHOLDER

kestrel-ridge-bearing-damage

Close-up of a large dismantled cylindrical roller bearing from a wind turbine gearbox on a workshop bench, the inner ring raceway showing a band of flaking and spalling damage, an engineer's gloved hand pointing with a pen, neutral workshop lighting.

Damaged ISS bearing after removal (photo: Highline Wind Services, February 2026)

2. Timeline

Date / time	Event
2025-11-04	Routine oil sample T-07: Fe 38 ppm, classified 'watch' by lab – no follow-up action
2025-12 (whole month)	CMS shows rising kurtosis on ISS bearing; below alarm threshold
2026-01-14 02:40	ISS bearing temperature starts rising (71 °C)
2026-01-14 03:42	Turbine trip, alarm received in control room
2026-01-14 09:15	Highline technicians on T-07, endoscope inspection
2026-01-14 14:30	Incident opened, Asset Operations Manager and HSE Manager informed
2026-01-19	Exchange of complete gearbox agreed with Highline under warranty (contract WE-CON-2023-015)
2026-01-29	Replacement gearbox delivered from Weserkraft Getriebe GmbH pool stock
2026-02-02 – 02-05	Crane works by Borders Heavy Lift Ltd: 750 t crawler crane, gearbox exchange
2026-02-06 14:20	T-07 returned to service after commissioning tests

3. Consequences

- No injuries. No oil released outside the nacelle (approx. 4 litres collected in the nacelle drip tray).
- Turbine offline for 23 days; estimated lost production 1,046 MWh (see lost production calculation).
- Site availability January 2026: 94.1 % (97.3 % excluding T-07); February 2026: 96.2 %.

- Repair costs (gearbox, crane, labour) borne by Highline Wind Services under the full-service agreement.

4. Immediate actions

- T-07 locked out and tagged; gearbox oil drained and sampled for laboratory analysis.
- CMS data of all 18 turbines reviewed on 15 January – no comparable trend found on other units.
- Crane lift plan and lift-specific risk assessment reviewed by HSE before mobilisation (winter conditions, crane pad bearing capacity).
- Notification to insurer (machinery breakdown) as a precaution on 16 January.

5. Follow-up

A formal root cause analysis has been commissioned with Highline and the gearbox manufacturer (teardown at the manufacturer's workshop in Bremen). Lost production will be claimed against the availability guarantee and warranty provisions of WE-CON-2023-015 as far as contractually possible.

Role	Name	Signed
Asset Operations Manager	David Chen	21 Jan 2026
Site Manager	Ben Okoye	21 Jan 2026
HSE Manager	Hannah Weber	21 Jan 2026